# Week 4: Selected Problems and Solutions

1. According to a July 31, 2013, posting on cnn.com subsequent to the death of a child who bit into a peanut, a 2010 study in the journal Pediatrics found that 8% of children younger than 18 in the United States have at least one food allergy. Among those with food allergies, about 39% had a history of severe reaction.

   (a) If a child younger than 18 is randomly selected, what is the probability that he or she has at least one food allergy and a history of severe reaction?

   (b) It was also reported that 30% of those with an allergy in fact are allergic to multiple foods. If a child younger than 18 is randomly selected, what is the probability that he or she is allergic to multiple foods?

   Solution:

   **a.** Let $A$ = child has a food allergy, and $R$ = child has a history of severe reaction. We are told that $P(A) = .08$ and $P(R \mid A) = .39$. By the multiplication rule, $P(A \cap R) = P(A) \times P(R \mid A) = (.08)(.39) = .0312$.

   **b.** Let $M$ = the child is allergic to multiple foods. We are told that $P(M \mid A) = .30$, and the goal is to find $P(M)$. But notice that $M$ is actually a subset of $A$: you can't have multiple food allergies without having at least one such allergy! So, apply the multiplication rule again:
   $P(M) = P(M \cap A) = P(A) \times P(M \mid A) = (.08)(.30) = .024$.

2. At a certain gas station, 40% of the customers use regular gas ($A_1$), 35% use plus gas ($A_2$), and 25% use premium ($A_3$). Of those customers using regular gas, only 30% fill their tanks (event B). Of those customers using plus, 60% fill their tanks, whereas of those using premium, 50% fill their tanks.

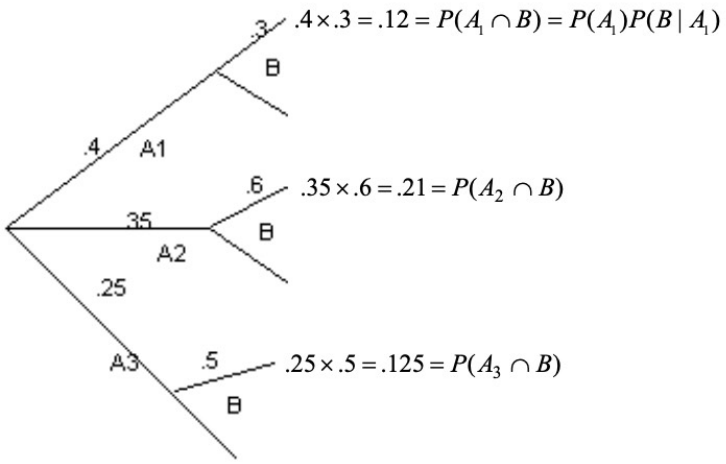
   (a) What is the probability that the next customer will request plus gas and fill the tank ($A_2 \cap B$)?

   (b) What is the probability that the next customer fills the tank?

   (c) If the next customer fills the tank, what is the probability that regular gas is requested? Plus? Premium?

   Solution:

   The required probabilities appear in the tree diagram below.




   **a.** $P(A_2 \cap B) = .21$.

   **b.** By the law of total probability, $P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B) = .455$.

   **c.** Using Bayes' theorem, $P(A_1 \mid B) = \dfrac{P(A_1 \cap B)}{P(B)} = \dfrac{.12}{.455} = .264$ ; $P(A_2 \mid B) = \dfrac{.21}{.455} = .462$ ; $P(A_3 \mid B) = 1 - .264 - .462 = .274$. Notice the three probabilities sum to 1.

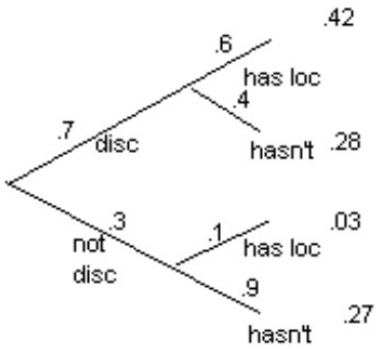
3. Seventy percent of the light aircraft that disappear while in flight in a certain country are subsequently discovered. Of the aircraft that are discovered, 60% have an emergency locator, whereas 90% of the aircraft not discovered do not have such a locator. Suppose a light aircraft has disappeared.

   (a) If it has an emergency locator, what is the probability that it will not be discovered?

   (b) If it does not have an emergency locator, what is the probability that it will be discovered?

   Solution:

   The tree diagram below shows the probability for the four disjoint options; e.g., $P$(the flight is discovered and has a locator) = $P$(discovered)$P$(locator | discovered) = (.7)(.6) = .42.




   **a.** $P(\text{not discovered} \mid \text{has locator}) = \dfrac{P(\text{not discovered} \cap \text{has locator})}{P(\text{has locator})} = \dfrac{.03}{.03+.42} = .067$.

   **b.** $P(\text{discovered} \mid \text{no locator}) = \dfrac{P(\text{discovered} \cap \text{no locator})}{P(\text{no locator})} = \dfrac{.28}{.55} = .509$.

4. For customers purchasing a refrigerator at a certain appliance store, let $A$ be the event that the refrigerator was manufactured in the U.S., $B$ be the event that the refrigerator had an icemaker, and $C$ be the event that the customer purchased an extended warranty. Relevant probabilities are:

   $P(A) = .75$ $P(B|A) = .9$ $P(B|A') = .8$ $P(C|A \cap B) = .8$ $P(C|A \cap B') = .6$ $P(C|A' \cap B) = .7$ $P(C|A' \cap B') = .3$

   (a) Construct a tree diagram consisting of first-, second-, and third-generation branches, and place an event label and appropriate probability next to each branch.

   (b) Compute $P(A \cap B \cap C)$.

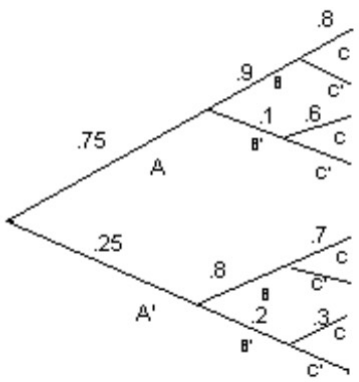
   (c) Compute $P(B \cap C)$.

   (d) Compute $P(C)$.

   (e) Compute $P(A|B \cap C)$, the probability of a U.S. purchase given that an icemaker and extended warranty are also purchased.

   Solution:

   **a.**



**b.** From the top path of the tree diagram, $P(A \cap B \cap C) = (.75)(.9)(.8) = .54$.

**c.** Event $B \cap C$ occurs twice on the diagram: $P(B \cap C) = P(A \cap B \cap C) + P(A' \cap B \cap C) = .54 + (.25)(.8)(.7) = .68$.

**d.** $P(C) = P(A \cap B \cap C) + P(A' \cap B \cap C) + P(A \cap B' \cap C) + P(A' \cap B' \cap C) = .54 + .045 + .14 + .015 = .74$.

**e.** Rewrite the conditional probability first: $P(A \mid B \cap C) = \dfrac{P(A \cap B \cap C)}{P(B \cap C)} = \dfrac{.54}{.68} = .7941$.

5. An oil exploration company currently has two active projects, one in Asia and the other in Europe. Let A be the event that the Asian project is successful and B be the event that the European project is successful. Suppose that A and B are independent events with $P(A) = .4$ and $P(B) = .7$.

   (a) If the Asian project is not successful, what is the probability that the European project is also not successful? Explain your reasoning.

   (b) What is the probability that at least one of the two projects will be successful?

   Solution:

   **a.** Since the events are independent, then $A'$ and $B'$ are independent, too. (See the paragraph below Equation 2.7.) Thus, $P(B'|A') = P(B') = 1 - .7 = .3$.

   **b.** Using the addition rule, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = .4 + .7 - (.4)(.7) = .82$. Since $A$ and $B$ are independent, we are permitted to write $P(A \cap B) = P(A)P(B) = (.4)(.7)$.

6. Let $X$ be the number of students who show up for a professor's office hour on a particular day. Suppose that the pmf of $X$ is given as $p(0) = 0.20$, $p(1) = 0.25$, $p(2) = 0.30$, $p(3) = 0.15$, and $p(4) = 0.10$.

   (a) Draw the corresponding probability histogram.

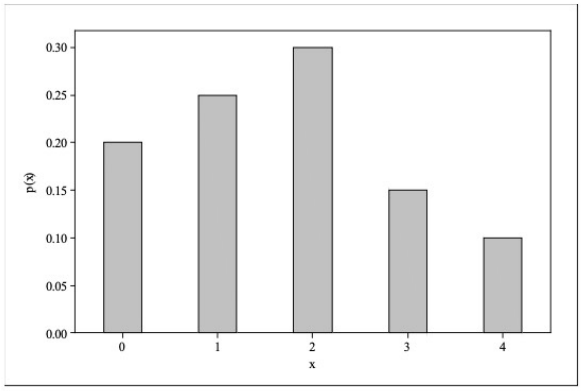
   (b) What is the probability that at least two students show up? More than two students show up?

   (c) What is the probability that between one and three students, inclusive, show up?

   (d) What is the probability that the professor shows up?

   Solution:

   **a.**




   **b.** $P(X \geq 2) = p(2) + p(3) + p(4) = .30 + .15 + .10 = .55$, while $P(X > 2) = .15 + .10 = .25$.

   **c.** $P(1 \leq X \leq 3) = p(1) + p(2) + p(3) = .25 + .30 + .15 = .70$.

   **d.** Who knows? (This is just a little joke by the author.)

7. Airlines sometimes overbook flights. Suppose that for a plane with 50 seats, 55 passengers have tickets. Define the random variable $Y$ as the number of ticketed passengers who actually show up for the flight. The probability mass function of $Y$ is given in the accompanying table:

| $y$ | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $p(y)$ | 0.05 | 0.10 | 0.12 | 0.14 | 0.25 | 0.17 | 0.06 | 0.05 | 0.03 | 0.02 | 0.01 |

(a) What is the probability that the flight will accommodate all ticketed passengers who show up?

(b) What is the probability that not all ticketed passengers who show up can be accommodated?

(c) If you are the first person on the standby list (the first one to get on the plane if there are any available seats), what is the probability that you will be able to take the flight? What is this probability if you are the third person on the standby list?

Solution:

**a.** Since there are 50 seats, the flight will accommodate all ticketed passengers who show up as long as there are no more than 50. $P(Y \leq 50) = .05 + .10 + .12 + .14 + .25 + .17 = .83$.

**b.** This is the complement of part **a**: $P(Y > 50) = 1 - P(Y \leq 50) = 1 - .83 = .17$.

**c.** If you're the first standby passenger, you need no more than 49 people to show up (so that there's space left for you). $P(Y \leq 49) = .05 + .10 + .12 + .14 + .25 = .66$. On the other hand, if you're <u>third</u> on the standby list, you need no more than 47 people to show up (so that, even with the two standby passengers ahead of you, there's still room). $P(Y \leq 47) = .05 + .10 + .12 = .27$.